SYAMPRASAD KALAVOOR

Python Developer | Computer Vision | AI/ML

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PROFESSIONAL SUMMARY

Python Developer with 4+ years of experience building and deploying real-time computer vision, machine learning, video analytics, and web applications. Hands-on experience with YOLO, OpenCV, TensorRT, NVIDIA Jetson, PyTorch, TensorFlow, FAISS, REST APIs, and production-oriented inference pipelines. Recently developed a local GenAI job-agent system using Qwen3:8B and Ollama for resume/JD analysis, ATS optimization, CV tailoring, validation, and automated document generation.

CORE TECHNOLOGY

Languages	Python, C, C++, JavaScript, Java, PHP, HTML
AI / Computer Vision	OpenCV, YOLO/YOLOv8, PyTorch, TensorFlow, Keras, ONNX Runtime, FAISS, face_recognition, Scikit-learn
Backend / APIs	Flask, Django, REST APIs, Postman, FFMPEG
GenAI / NLP	Qwen3:8B, Ollama, Prompt Engineering, NLP, NLTK, spaCy, TextBlob, ATS Analysis
Data / Databases	MySQL, MS SQL, ClickHouse, SQLite, NumPy, Pandas, Matplotlib
Edge / Deployment	NVIDIA Jetson AGX, Jetson Nano, TensorRT, ONNX Deployment, Embedded Systems, Real-time Processing
Engineering	Git, Multiprocessing, Threading, Subprocess, Schedule, Arduino

PROFESSIONAL EXPERIENCE

Freelance Developer & Technical Mentor | May 2025 – Present

Freelance / Independent

- Delivered Python, machine learning and computer-vision projects based on project requirements.
- Conducted technical mentoring and practical workshops covering Python, ML and project development.

Python Developer – Video Analytics & ML | Sep 2022 – May 2025

GAMA Security Solution, Dubai

- Developed and deployed real-time video analytics solutions using Python, OpenCV, YOLO and optimized inference pipelines.
- Worked on edge AI deployments using NVIDIA Jetson and TensorRT for continuous RTSP stream processing.
- Implemented detection, tracking, occupancy, face analytics, alerts and database/API integrations for production systems.

Python Developer – Video Analytics, ML & Web | May 2021 – May 2022

Nacre Systems, Palakkad

- Built computer-vision and web applications using Python, OpenCV, ML frameworks and REST-based integrations.
- Worked on NLP/chatbot functionality and data-driven application components.

Python Developer – Video Analytics, ML & Web | Nov 2020 – May 2021

Grapesgenix Technical Solutions, Thrissur

- Developed Python-based computer-vision and web solutions and supported project mentoring/training activities.

SELECTED PROJECTS

AI Job Agent – Intelligent Resume & Job Matching System

Developed a Python-based AI system that analyzes resumes against job descriptions, calculates match/ATS scores, identifies skill gaps and ATS keywords, generates tailored CV content using locally hosted Qwen3:8B via Ollama, validates generated content against the source resume, and produces DOCX/PDF outputs.

Tools: Python, Qwen3:8B, Ollama, NLP, Prompt Engineering, ATS Analysis, PyMuPDF, python-docx

Smart Surveillance System – NVIDIA Jetson AGX

Built an edge AI real-time surveillance platform with YOLO-based people/vehicle detection, tracking and ID assignment, line-crossing entry/exit analytics, live heatmaps, TensorRT optimization and continuous RTSP stream processing.

Tools: Python, OpenCV, YOLO, TensorRT, NVIDIA Jetson, RTSP

Real-Time Face Recognition & Face Search

Developed biometric systems for real-time face recognition and image-based face search using facial embeddings and FAISS indexing for fast top-k similarity retrieval from large datasets.

Tools: Python, OpenCV, face_recognition, FAISS, Deep Learning

ANPR with OCR

Developed an automatic number plate recognition pipeline for real-time camera streams using computer vision, deep learning and OCR to detect and read vehicle license plates.

Tools: Python, OpenCV, YOLO, OCR, Real-time Video

RTSP Analytics – In/Out, Occupancy & Fence Crossing

Implemented real-time person counting, room occupancy limits and perimeter/fence-crossing alerts using YOLO detection, tracking and line-crossing logic over RTSP camera streams.

Tools: Python, YOLO, OpenCV, RTSP, Tracking

Fall Detection & PPE Monitoring

Built real-time safety analytics for fall detection using YOLOv8 pose estimation and PPE/safety-dress detection for industrial monitoring and alerting.

Tools: Python, YOLOv8 Pose, OpenCV, Computer Vision

Nail Segmentation for Virtual Try-On

Implemented YOLO-based image segmentation to isolate fingernails for virtual nail try-on applications.

Tools: Python, YOLO, Image Segmentation, Computer Vision

Smoke Detection System – Arduino + MQ Sensor

Designed a real-time smoke/gas monitoring system using Arduino and MQ sensors with threshold-based alerting for safety applications.

Tools: Arduino, Sensors, Python/Embedded Integration

EDUCATION

Bachelor of Computer Applications (BCA) | 2018 – 2021

Python Django Full Stack Course | 2020

LANGUAGES & WEB CV

Malayalam • English • Tamil • Hindi

Know more about my projects: https://syamprasadkalavoor.github.io/cv/